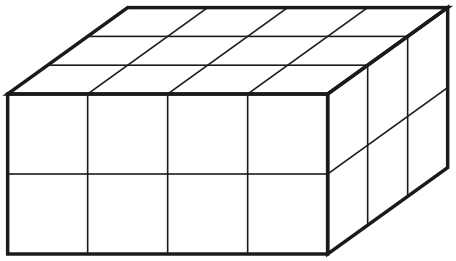
Volume

How many unit cubes fill the box? Count one layer, then the layers.

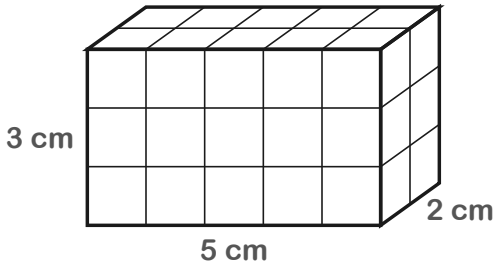
A box 4 cubes long, 3 cubes wide, 2 cubes tall



One layer: 4 rows of 3 = 12 cubes.
2 layers: 12 + 12 = 24 cubes.
Volume is how many cubes fill it.

4 × 3 × 2 = 24 cubes

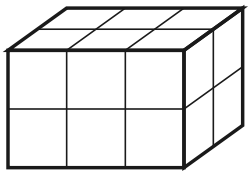
Length × width × height



Each cube is 1 cm on every edge: a cubic cm.
Base: 5 × 2 = 10 squares. Height: 3 layers.

5 × 2 × 3 = 30 cubic cm

Your turn!



volume = 12 cubes

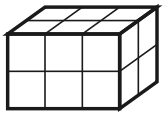
For teachers

Goal: volume as counting unit cubes, then length × width × height (5.MD.3–5). Say "How many cubes in one layer? How many layers?" Units are cubic: cubic cm, cubic m.

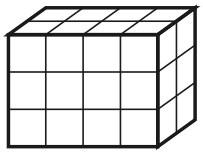
Volume: Practice

One layer, then the layers. Length × width × height.

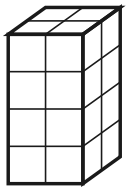
Part 1 Count the cubes Some cubes are hidden. Count one layer, then the layers.

1) 

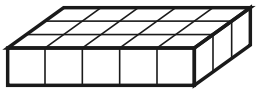
12 cubes

2) 

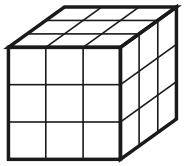
24 cubes

3) 

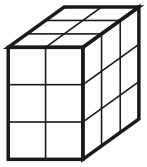
16 cubes

4) 

15 cubes

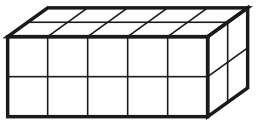
5) 

27 cubes

6) 

18 cubes

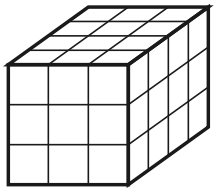
Part 2 Layers Cubes in one layer, then how many layers.

7) 

one layer: 5 × 2 = 10

layers: 2

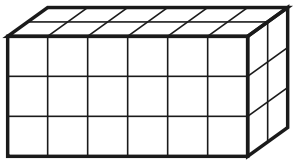
10 × 2 = 20 cubes

8) 

one layer: 3 × 4 = 12

layers: 3

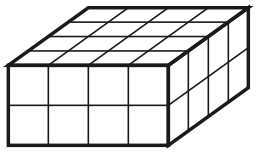
12 × 3 = 36 cubes

9) 

one layer: 6 × 2 = 12

layers: 3

12 × 3 = 36 cubes

10) 

one layer: 4 × 4 = 16

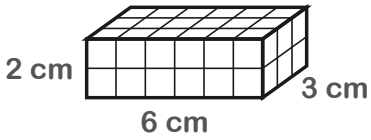
layers: 2

16 × 2 = 32 cubes

Volume: Practice (continued)

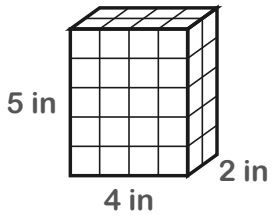
Part 3 Length × width × height Use the numbers on the edges.

11)



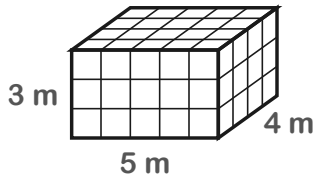
$6 \times 3 \times 2 =$ 36 cubic cm

12)



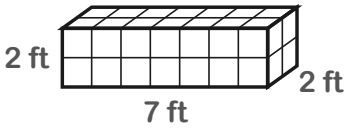
$4 \times 2 \times 5 =$ 40 cubic in

13)



$5 \times 4 \times 3 =$ 60 cubic m

14)



$7 \times 2 \times 2 =$ 28 cubic ft

Part 4 Find the volume No help this time.

15) 8 cm × 3 cm × 2 cm = 48 cubic cm

16) 5 in × 5 in × 4 in = 100 cubic in

17) 9 m × 2 m × 3 m = 54 cubic m

18) 6 cm × 4 cm × 5 cm = 120 cubic cm

19) 10 ft × 3 ft × 2 ft = 60 cubic ft

20) 7 cm × 5 cm × 2 cm = 70 cubic cm

Volume: Practice (continued)

Part 4 Find the volume (continued)

21) 4 in × 4 in × 4 in = 64 cubic in

22) 12 m × 2 m × 3 m = 72 cubic m

23) 7 ft × 6 ft × 2 ft = 84 cubic ft

24) 9 cm × 4 cm × 3 cm = 108 cubic cm

Part 5 Base times height The base is already worked out.

25) 20 sq cm × 3 cm = 60 cubic cm

26) 15 sq in × 5 in = 75 cubic in

27) 36 sq m × 2 m = 72 cubic m

28) 24 sq cm × 5 cm = 120 cubic cm

29) 18 sq ft × 6 ft = 108 cubic ft

30) 40 sq in × 2 in = 80 cubic in

31) 12 sq m × 7 m = 84 cubic m

32) 50 sq cm × 4 cm = 200 cubic cm

Volume: Practice (continued)

Part 6 Find the missing edge Volume ÷ the other two edges.

33) $4 \times 5 \times \boxed{3} = 60$

34) $6 \times 2 \times \boxed{7} = 84$

35) $3 \times 3 \times \boxed{5} = 45$

36) $8 \times 4 \times \boxed{2} = 64$

37) $5 \times 6 \times \boxed{4} = 120$

38) $7 \times 3 \times \boxed{6} = 126$

Part 7 Two boxes No help this time. Find each volume, then add.

39) $3 \times 2 \times 2 + 4 \times 2 \times 1 = \boxed{12} + \boxed{8} = \boxed{20}$

40) $5 \times 3 \times 2 + 2 \times 2 \times 3 = \boxed{30} + \boxed{12} = \boxed{42}$

41) $4 \times 4 \times 3 + 6 \times 2 \times 1 = \boxed{48} + \boxed{12} = \boxed{60}$

42) $6 \times 3 \times 2 + 3 \times 3 \times 3 = \boxed{36} + \boxed{27} = \boxed{63}$

Volume: Practice (continued)

Part 8 Find the volume

43) 10 cm × 8 cm × 4 cm = 320 cubic cm

44) 8 m × 4 m × 6 m = 192 cubic m

45) 7 cm × 4 cm × 5 cm = 140 cubic cm

46) 10 m × 5 m × 6 m = 300 cubic m

47) 2 ft × 8 ft × 4 ft = 64 cubic ft

48) 9 ft × 7 ft × 2 ft = 126 cubic ft

49) 2 cm × 5 cm × 4 cm = 40 cubic cm

50) 11 cm × 7 cm × 4 cm = 308 cubic cm

51) 2 in × 2 in × 5 in = 20 cubic in

52) 5 cm × 3 cm × 3 cm = 45 cubic cm

53) 2 ft × 3 ft × 6 ft = 36 cubic ft

54) 5 in × 4 in × 3 in = 60 cubic in

Volume: Practice (continued)

Part 8 Find the volume (continued)

55) 11 ft × 5 ft × 5 ft = 275 cubic ft

56) 9 ft × 3 ft × 5 ft = 135 cubic ft

57) 3 ft × 3 ft × 2 ft = 18 cubic ft

58) 9 ft × 6 ft × 5 ft = 270 cubic ft

59) 3 in × 5 in × 4 in = 60 cubic in

60) 9 ft × 4 ft × 6 ft = 216 cubic ft

Part 9 Find the missing edge

61) 7 × 4 × 8 = 224

62) 4 × 8 × 5 = 160

63) 7 × 5 × 2 = 70

64) 8 × 2 × 2 = 32

65) 2 × 3 × 7 = 42

66) 8 × 9 × 2 = 144

Volume: Practice (continued)

Part 9 Find the missing edge (continued)

67) $4 \times 2 \times \boxed{3} = 24$

68) $7 \times 4 \times \boxed{4} = 112$

69) $2 \times 4 \times \boxed{8} = 64$

70) $8 \times 4 \times \boxed{9} = 288$

71) $4 \times 9 \times \boxed{8} = 288$

72) $5 \times 5 \times \boxed{8} = 200$

73) $9 \times 9 \times \boxed{8} = 648$

74) $7 \times 7 \times \boxed{3} = 147$

75) $6 \times 9 \times \boxed{4} = 216$

76) $4 \times 6 \times \boxed{3} = 72$

77) $6 \times 2 \times \boxed{9} = 108$

78) $6 \times 8 \times \boxed{2} = 96$

Volume: Practice (continued)

Part 10 Remember? Problems from earlier packets.

79) $\frac{2}{3} \times \frac{3}{5} = \frac{6}{15}$

80) $\frac{3}{4} \times 16 = 12$

81) $\frac{1}{2} + \frac{2}{5} = \frac{9}{10}$

82) $8 \times 0.45 = 3.6$

83) $6.5 - 2.75 = 3.75$

84) $38 \times 25 = 950$

Part 11 Word problems Length × width × height.

85) A fish tank is 6 ft long, 2 ft wide and 3 ft tall. What is its volume?

Number sentence $6 \times 2 \times 3 = 36$

Answer 36 cubic ft

Draw or show your work

86) A box holds 5 layers of cubes. Each layer has 12 cubes. How many cubes fill the box?

Number sentence $12 \times 5 = 60$

Answer 60 cubes

Draw or show your work

87) A sandbox is 4 m long and 3 m wide. Sand fills it 1 m deep. How much sand is that?

Number sentence $4 \times 3 \times 1 = 12$

Answer 12 cubic m

Draw or show your work

Volume: Practice (continued)

Part 11 Word problems (continued)

88) A box has a volume of 80 cubic cm. Its base is 4 cm by 5 cm. How tall is it?

Number sentence $4 \times 5 \times 4 = 80$

Answer **4** cm

Draw or show your work

89) Challenge: Two boxes are stacked. One is 5 × 4 × 2 cm, the other 3 × 3 × 3 cm. What is the volume in all?

Number sentence $40 + 27 = 67$

Answer **67** cubic cm

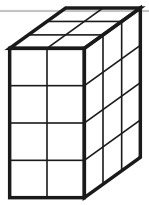
Draw or show your work

Check-In: Volume

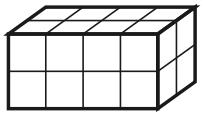
Show what you know. No help this time.

Mastery: 7 of 8 (80%)

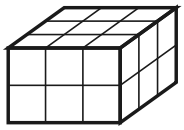
Part 1 Count the cubes

1) 

24 cubes

2) 

16 cubes

3) 

18 cubes

Part 2 Find the volume

4) 8 m × 2 m × 5 m = 80 cubic m

5) 5 in × 4 in × 4 in = 80 cubic in

6) 7 cm × 3 cm × 2 cm = 42 cubic cm

Part 3 Find the missing edge

7) 6 × 4 × 3 = 72

8) 2 × 7 × 5 = 70